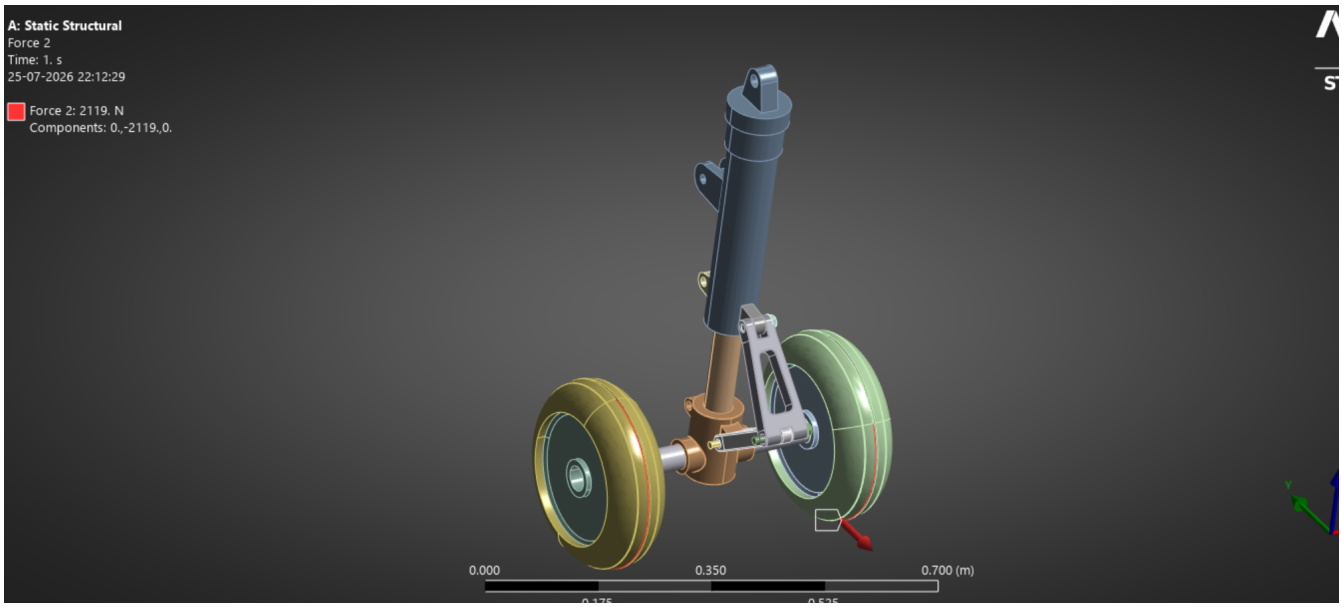
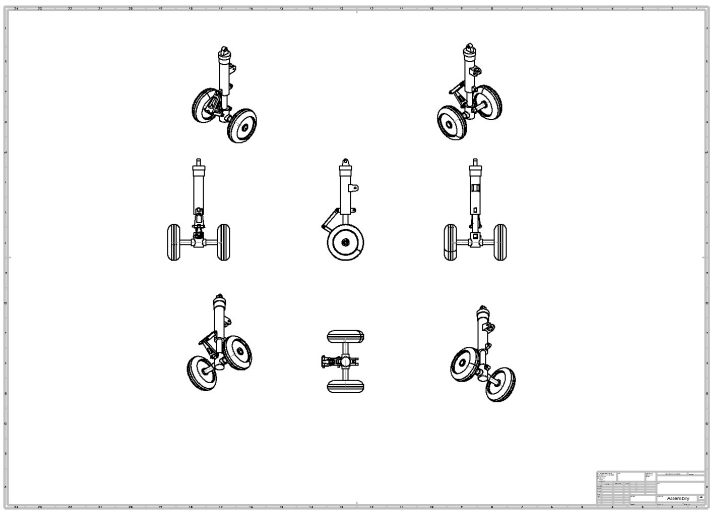
Complex assembly: twin-wheel nose landing gear

12 October 2026. Three original source families collected: landing gear, chambered finger and connecting rod. Original CAD and published FEM evidence retained. No locally generated geometry, solver runs or frontier-model calls.



Original publisher CAD illustration: assembled nose gear, source report physical page 1, Figure 1. Downloaded STEP, native SolidWorks assembly and seven source part files provide actual CAD assets.

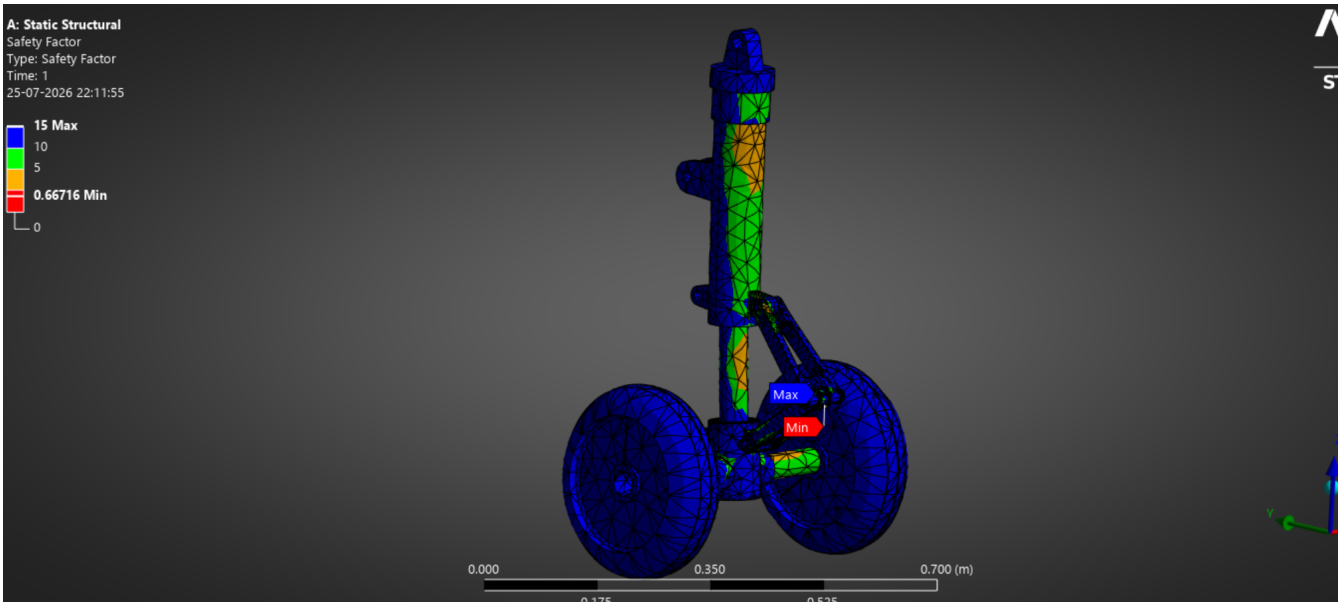


Original drawing: publisher orthographic/isometric assembly sheet. Original DXF and SLDDRW retained. The inspected DXF has zero DIMENSION entities and the raster has no visible dimension callouts, so a complete dimensioned engineering drawing is not claimed.

Independent STEP audit: valid shape, 15 solids, 500 faces; approximately 513.33 x 346.95 x 842.21 mm. Component instances, including wheel bodies and fasteners, are not counted as separate FEM tasks.

Source: Aryan Pandharipande, aircraft nose landing gear design. MIT source license, original files and pinned revision/download hashes retained.

Landing gear: original FEM scalar references



Original FEM figure: static safety-factor contour, source report physical page 4, Figure 11. Image minimum 0.66716; report text rounds it to 0.667. Source identifies the upper torque-link fillet as the critical location.

Reference quantity	Original value / provenance
Minimum static safety factor	0.66716, original Figure 11 legend
Average static safety factor	13.63, physical page 5 Table II
Maximum plotted safety factor	15, explicitly capped. Not an exact uncensored strength target.

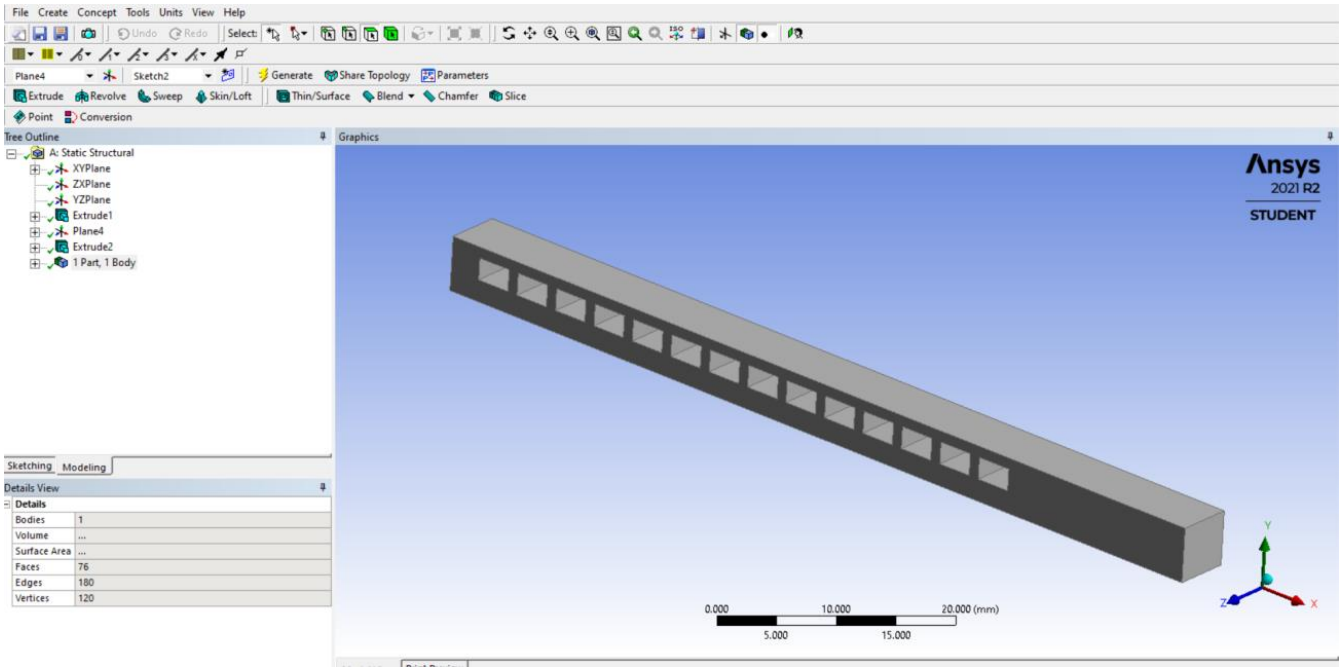
Simultaneous source loads	Magnitude and scope
Axle vertical force	7062 N, +Z, two axle edges
Lateral force	2119 N, -Y, two axle edges
Piston axial force	15696 N, +Z, one piston face

Saved-output coverage: only a standalone .wbpj project pointer is published. Dependent Mechanical files, mesh and raw .rst are absent. The numerical references are the report table and original contour, not a saved-field archive.

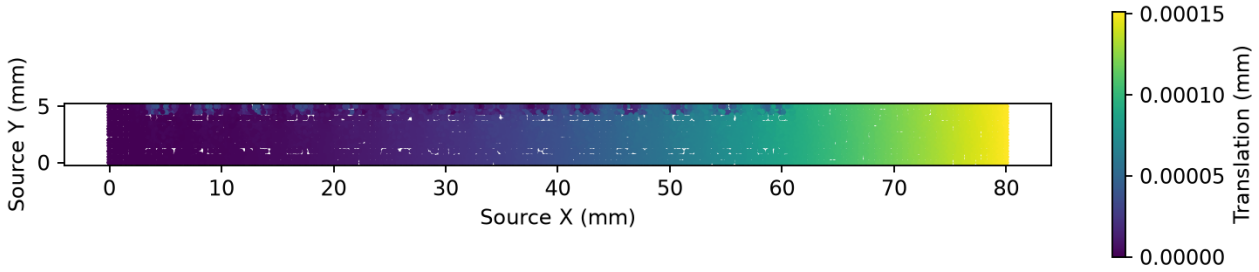
Admission limits: source uses baseline structural-steel assignments and a Nickel wheel hub. Contacts, actual material scope, mesh convergence and dynamic landing behavior require review. The reference model has safety factor below unity under its source assumptions. Repository mentions of fatigue and buckling do not establish verified result cases.

One native CAD/report-reference candidate. CAD creation from the supplied drawing and FEM from supplied CAD remain separate benchmark tracks, both pending complete reference setup.

Chambered finger: native CAD + saved solutions



Original publisher CAD screenshot: single-row chambered finger, source report physical page 2. Each downloaded source project includes a native DesignModeler SYS.agdb with serialized Parasolid/object data, input ds.dat, Mechanical database and file.rst. Native BREP validity has not been independently imported; no original neutral STEP found.



Read-only reference plot: saved Part2_1 displacement magnitude at original undeformed mesh coordinates. Every fourth point displayed for legibility; full node arrays retained. These are the stored source solution values, read without launching ANSYS.

Saved project	Nodes / elements	Maximum translation mm
Part1	73,218 / 18,933	1.249597696e-05
Part2_1	81,041 / 21,415	0.0001510689569
Part2_2	81,041 / 21,381	0.0001511545658

Material scope: report describes a PDMS finger, but all three saved input decks use E = 200,000 MPa, Poisson ratio 0.3 and steel-like density. Pressure is 0.18 MPa. The saved runs are linear static baselines, not verified PDMS or hyperelastic reference simulations. Preserve this conflict.

Source: electronjia, ANSYS finger project. One geometry family, three saved solutions. Source-data license unresolved; files retained locally for research. No public redistribution performed.

Saved field interpretation and scoring limits

Saved project	Finite stress rows / total nodes	Finite-subset peak VM MPa
Part1	19,003 / 73,218	1.343744338
Part2_1	20,417 / 81,041	3.103679765
Part2_2	20,417 / 81,041	2.917912315

Displacement: all recorded translations are finite and node IDs are unique. Each NPZ retains the reader-returned displacement arrays, original mesh coordinates and node IDs. Every NPZ was compared back to the reader arrays, with NaNs preserved.

Stress recovery: ansys-mapdl-reader 0.56.0 nodal_stress averages stored element-nodal stresses at common corner nodes. It is a recovered averaged field, rather than an unaveraged integration-point field. Component order is X,Y,Z,XY,YZ,XZ. The table maximum is computed from the six finite recovered components.

The reader returns missing stress entries at 54,215 nodes for Part1 and 60,624 nodes for each Part2 model. Missing output is consistent with the corner-node convention; exact element/output support still requires review. Never replace these NaNs with zero or score them as zero stress.

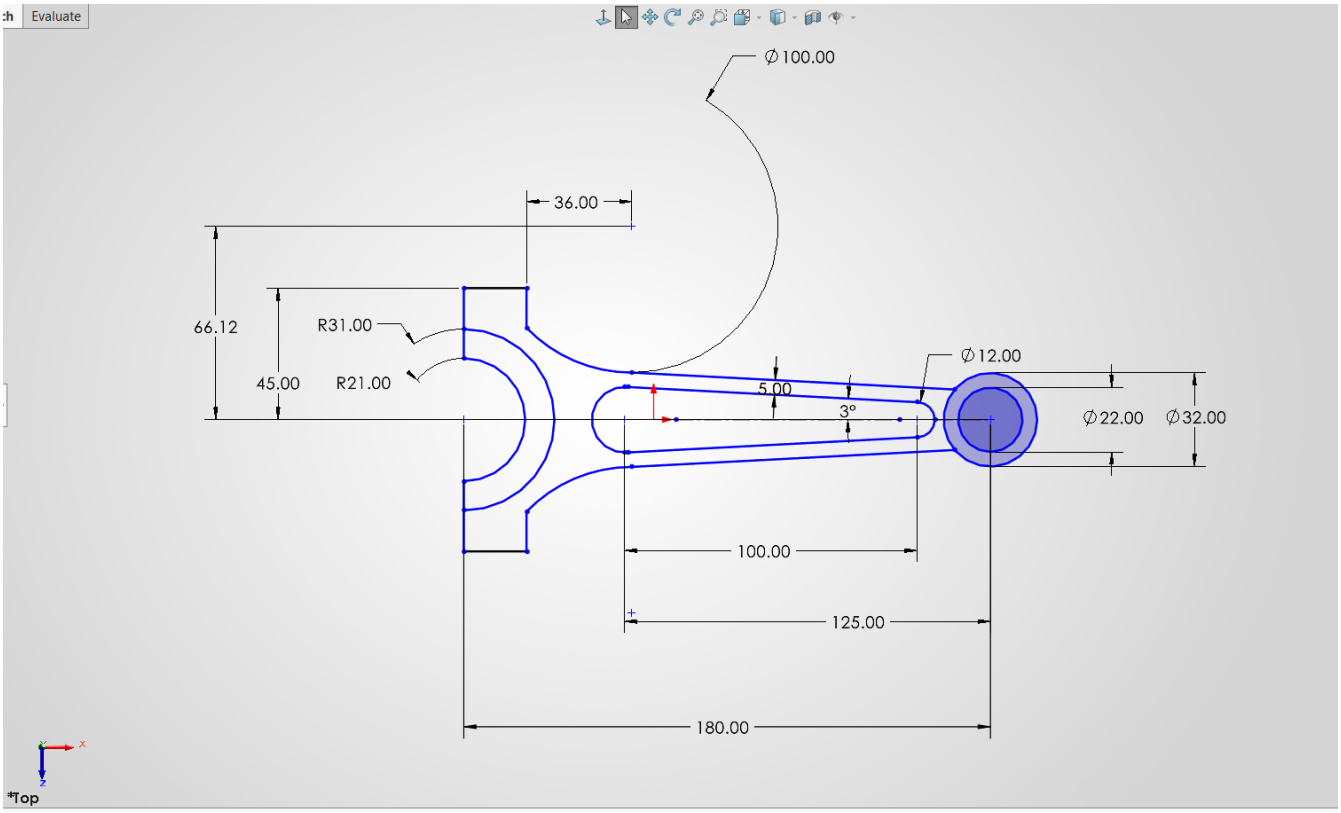
Units and solution state: binary header says User Defined, while the source input explicitly declares /units,MPA and consistent N-mm units. Three solutions have one static set each, labeled 1. This label does not establish a transient time series. Source logs report completion, two warnings and zero errors; this alone does not establish correct physical modeling.

Potential metric	Required interpretation
Displacement error	Match the declared source material, pressure, support constraints, node mapping and units. Do not compare with a different PDMS model.
Recovered stress error	Use matching corner-node averaging and a finite reference mask. Report coverage. Do not compare directly against unaveraged element/integration-point stress.
Model accuracy scores	Not measured. No frontier-model trial or new FEM solve performed.

Reader methodology: official PyMAPDL Legacy Reader documentation. Same averaging convention applies to the earlier marine-anchor reader extraction; the collection index records this shared scoring note.

These are three native CAD/saved-solution candidates with explicit source-physics limits. Zero newly certified benchmark admissions.

Connecting rod: CAD sketches and source units



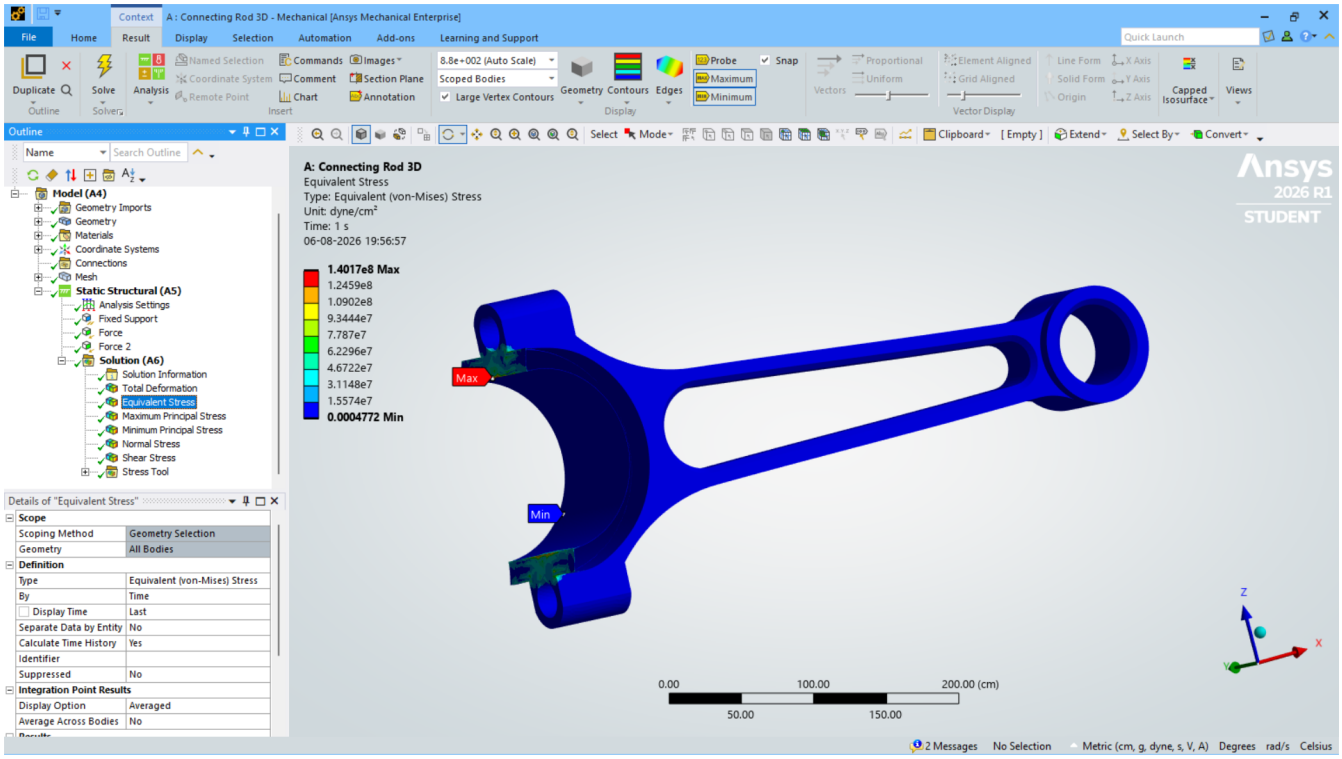
Original publisher CAD sketch: dimension labels from the native SolidWorks modeling workflow. Second sketch, final CAD view, STEP and SLDPRT retained. This is a dimensioned modeling sketch, not a verified manufacturing drawing with complete views, units, thickness and tolerances.

Original CAD audit	Result and correspondence limits
STEP shape	One valid solid, 63 faces. STEP explicitly declares conversion-based INCH.
Imported dimensions	4980.94 x 889.00 x 2286.00 mm. The maximum extent is 4.98 m. Source geometry dimension intent requires review.
Source FEM display	Screenshot uses cm and CGS stress units, with a 200 cm scale bar. No silent rescaling or new geometry applied.

Source static setup: report specifies steel E = 200 GPa, Poisson ratio 0.3, fixed big end and force components(-5000,0,-750)N at the small end. Only a standalone Workbench project pointer is available; no raw .rst or solver mesh/deck found.

Source: Pranav Mane, connecting rod design and analysis. Original README says educational/portfolio purposes; no standard license located. Local research inspection only, no public publication.

Connecting rod: FEM references with unit provenance



Original FEM screenshot: equivalent stress legend 1.4017e8 dyne/cm^2. Preserve the unusual units. One dyne/cm^2 equals 0.1 Pa, giving 14.017 MPa. Report text rounds the same quantity to 14.02 MPa.

Reference quantity	Original label	Explicit unit conversion
Maximum equivalent stress	1.4017e8 dyne/cm^2	14.017 MPa
Maximum total deformation	0.031969 cm	0.31969 mm
Displayed safety factor	15 minimum and maximum	Capped display; excluded as an exact uncensored target

The report gives 0.3197 mm deformation, consistent with rounding the screenshot value. Reference labels are taken from original plots; they do not provide nodal field errors. CAD scale, material scope, applied loads and convergence remain scoring checks.

Batch counts: three engineering families; three native finger CAD/saved-solution candidates, two STEP/plot-report candidates. Landing gear has 15 imported solids and rod has one; solids and load variants are not additional engineering families. No newly verified complete dimensioned engineering drawings or certified benchmark cases.

Files: original CAD, source drawings/sketches, reports, Workbench data, three raw RST files, three exact reader-output NPZs, source hashes, catalogs and inspection code. License scope is recorded per source. No COMSOL dependency. Reading saved RST files required no ANSYS solve; full replay still needs compatible tooling or validated translation.

Further leads: chiral kirigami CAD/nonlinear FEM archive identified, but download API requires a bearer token. The sliding-mechanism paper supplies data on request. CADEngBench has no verified downloadable reference repository in this search; the cited OpenAM Figshare collection API returned 404. None is included in collected pair counts.